

Experiment 010: H1 Presentation-Stress Report

Generated by run.py

1 Audit Data

This TeX file is a compact report generated from the same in-memory result object written to `results.json`. It is computational evidence only and does not prove presentation invariance.

2 Conventions

- Base field: $\mathbb{Q}$.
- Category: `nonunital Jordan algebras`.
- Target algebra: $J = k\{x, y\}/(x^2, xy, y^2)$.
- Matrix convention: source rows, target coordinates.
- Canonical relation basis: (x^2, xy, y^2) .
- Minimal relation cells: (r_{xx}, r_{xy}, r_{yy}) .

3 Outcome Summary

Case	Outcome	$\dim C_1$	$\dim H_1$	Cert.
A_minimal_presentation	pass_strong	3	3	yes
C_redundant_relation_without_syzygy	expected_fail_observed	4	4	no
D_redundant_relation_with_syzygy	pass_strong	4	3	yes
F_truncation_depth_diagnostic	pass_strong	4	3	yes
B_linear_change_g1	pass_strong	3	3	yes
B_linear_change_g2	pass_strong	3	3	yes
B_linear_change_g3	pass_strong	3	3	yes
E_multiple_redundant_relations_m1	pass_strong	4	3	yes
E_multiple_redundant_relations_m2	pass_strong	5	3	yes
E_multiple_redundant_relations_m3	pass_strong	6	3	yes
E_multiple_redundant_relations_m5	pass_strong	8	3	yes

4 MVP Diagnostics

- Minimal presentation: outcome `pass_strong`, $\dim H_1 = 3$.
- Naive redundant-relation control: outcome `expected_fail_observed`, $\dim H_1 = 4$. This is the planned false extra H_1 class.
- Repaired redundant relation: outcome `pass_strong`, $\text{rank } \partial_2^Q = 1$, $\dim H_1 = 3$.
- Minimum passed: `true`.

5 Extended Stress Tests

5.1 Linear Changes Of Generators

Case	Outcome	$\text{Sym}^2(g)$ check
B_linear_change_g1	pass_strong	yes
B_linear_change_g2	pass_strong	yes
B_linear_change_g3	pass_strong	yes

5.2 Multiple Redundant Relations

Case	$\text{rank } \partial_2^Q$	$\dim H_1$	Cert.
E_multiple_redundant_relations_m1	1	3	yes
E_multiple_redundant_relations_m2	2	3	yes
E_multiple_redundant_relations_m3	3	3	yes
E_multiple_redundant_relations_m5	5	3	yes

6 Truncation Depth

Presentation model	$N = 1$	$N = 2$	$N = 3$
minimal_presentation	3	3	3
redundant_without_syzygy	4	4	4
redundant_with_syzygy	4	3	3

7 Basis-Alignment Certificates

For redundant-relation cases, the strong certificate checks $\text{im}(\partial_2^Q) = \ker(\rho)$ exactly over $\mathbb{Q}$. The naive control intentionally omits the syzygy, so the kernel of ρ is not killed and the false extra H_1 class remains.

Case	$\dim C_2$	$\text{rank } \partial_2^Q$	$\text{im } \partial_2^Q = \ker \rho$
C_redundant_relation_without_syzygy	0	0	no
D_redundant_relation_with_syzygy	1	1	yes
E_multiple_redundant_relations_m1	1	1	yes
E_multiple_redundant_relations_m2	2	2	yes
E_multiple_redundant_relations_m3	3	3	yes
E_multiple_redundant_relations_m5	5	5	yes

8 Limitations

This is a bounded low-degree diagnostic for one square-zero example. It supports H_1 stability under the listed presentation stresses but does not prove presentation invariance.

9 Interpretation

Experiment 010 supports the stability of the computed H1 for the two-generator square-zero Jordan algebra under deterministic linear generator changes and controlled redundant-relation refinements. The expected-fail control reproduces the false extra H1 class that appears when a redundant relation is added without the corresponding syzygy. This supports, but does not prove, the identification of the stable H1 with $\text{Sym}^2(k^2)$ in this bounded example.